

2. DNS: HOW A NAME BECOMES AN ADDRESS

One resolver asks around, then remembers

AN EVERYDAY ANALOGY: THE RECEPTIONIST

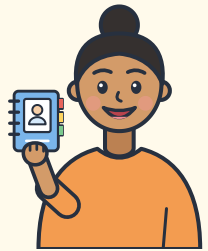


You

① Where is Example Bakery in Oakdale?

② 12 Mill Street. The bakery told me.

③ The next person gets it from my notes.

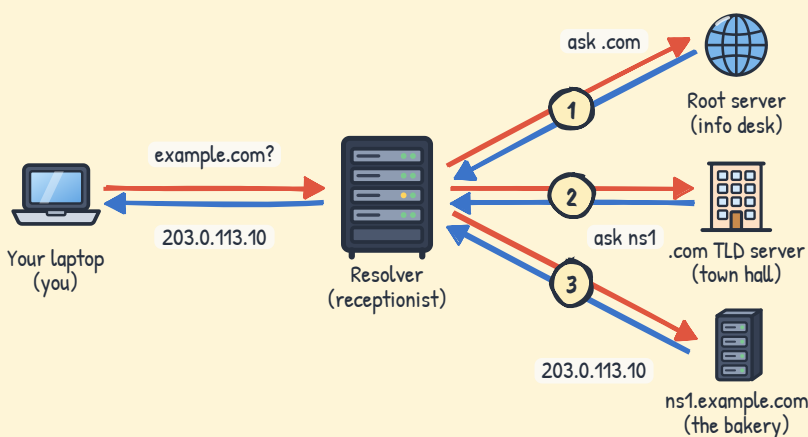


Receptionist

The receptionist follows the same routine every time:

- You ask one person. She does all the calling.
- She calls the info desk, town hall, then bakery.
- Only the bakery itself knows the address for sure.
- Her note is good for one hour, then she asks again.

THE SAME ROUTINE IN DNS



Root and TLD servers only point onward. The resolver makes every call.

HER NOTES: CACHE AND TTL



CACHE HIT

Same name again? The resolver answers from memory, with no calls at all.



TTL

Time to live, in seconds, set by the zone owner. 3600 is her one-hour note.



WHY CHANGES LAG

Change a record and old copies stay in caches until their TTL runs out.



MANY NOTEBOOKS

Your browser and operating system keep their own copies too.

WHAT THE BAKERY CAN ANSWER: RECORD TYPES



A

name → IPv4

www.example.com → 203.0.113.10



AAAA

name → IPv6

The same for IPv6, e.g. 2001:db8::10.



CNAME

alias

An alias for another name. It can hold no other record types.



MX

mail

Which servers take mail. The lower number is tried first.



TXT

free text

Notes for machines: SPF mail rules, domain ownership checks.



NS

name servers

The zone's authoritative servers. The TLD hands these out.

TRY IT IN A TERMINAL

```
$ dig example.com
Asks your resolver. The number after the name is the TTL left. Windows has nslookup for this.
```

```
$ dig +trace example.com
Walks root → TLD → authoritative by itself, like the receptionist.
```

PORT 53:

DNS servers listen on port 53, for both UDP and TCP.



- **UDP:** everyday lookups, one packet each way
- **TCP:** answers too big for UDP, and zone copies



REMEMBER: You ask one resolver. It asks root, TLD and the authoritative server, then keeps the answer until the TTL runs out.